

Q4

● Graded

Student

Steven Chen

Total Points

26.0334 / 31.5 pts

Question 1

(no title)

3 / 5 pts

+ 5 pts for each parent index starting from the last parent down to index 0:
bubbleDown(index)

+ 4.75 pts Very minor errors

+ 4.5 pts Minor errors

+ 4 pts Some errors

+ 3.5 pts Some errors

✓ + 3 pts Major errors

+ 2 pts Major errors

+ 1 pt Major errors

+ 0 pts Incorrect

Question 2

(no title)

1 / 1 pt

✓ + 1 pt 18

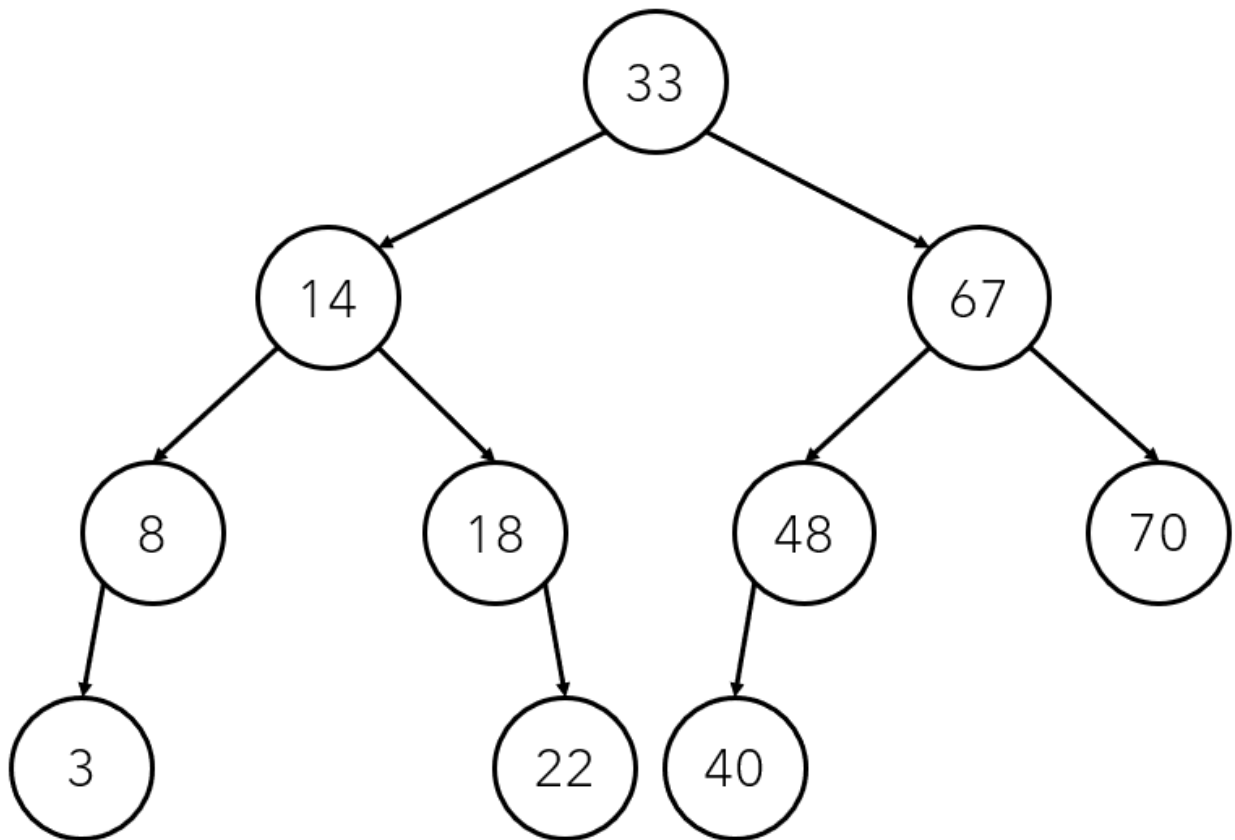
+ 0 pts Incorrect

Question 3

(no title)

2.6667 / 3 pts

+ 3 pts



✓ + 2.667 pts Minor errors

+ 2 pts Some errors

+ 1.75 pts Some errors

+ 0 pts Incorrect

Sorry, 20 did not need to be inserted ❤️ but you inserted and rotated it correctly except for miswriting 14 as 12

Question 4

(no title)

1 / 1 pt

✓ + 1 pt 2 or 3

+ 0 pts Incorrect

Question 5

(no title)

1 / 1 pt

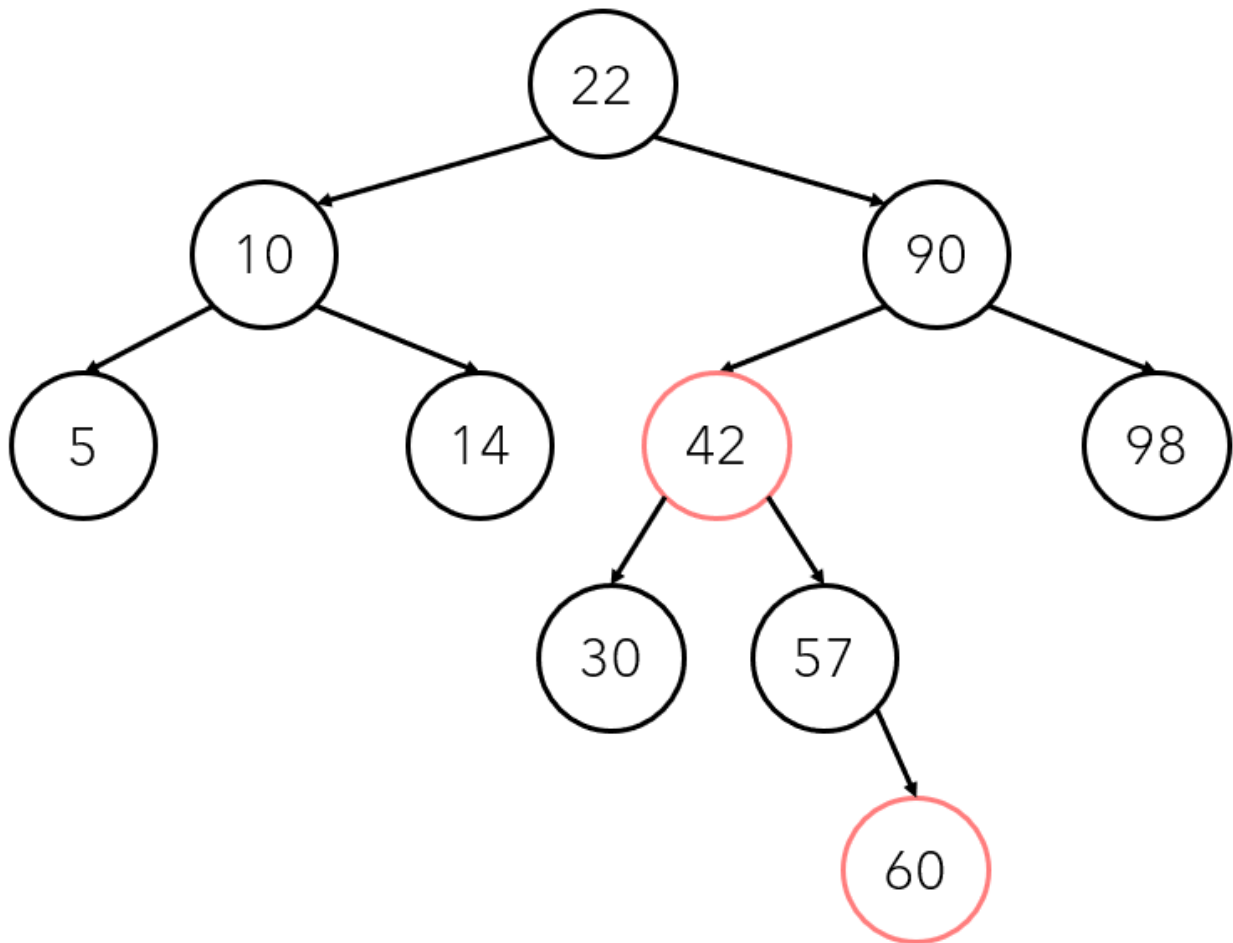
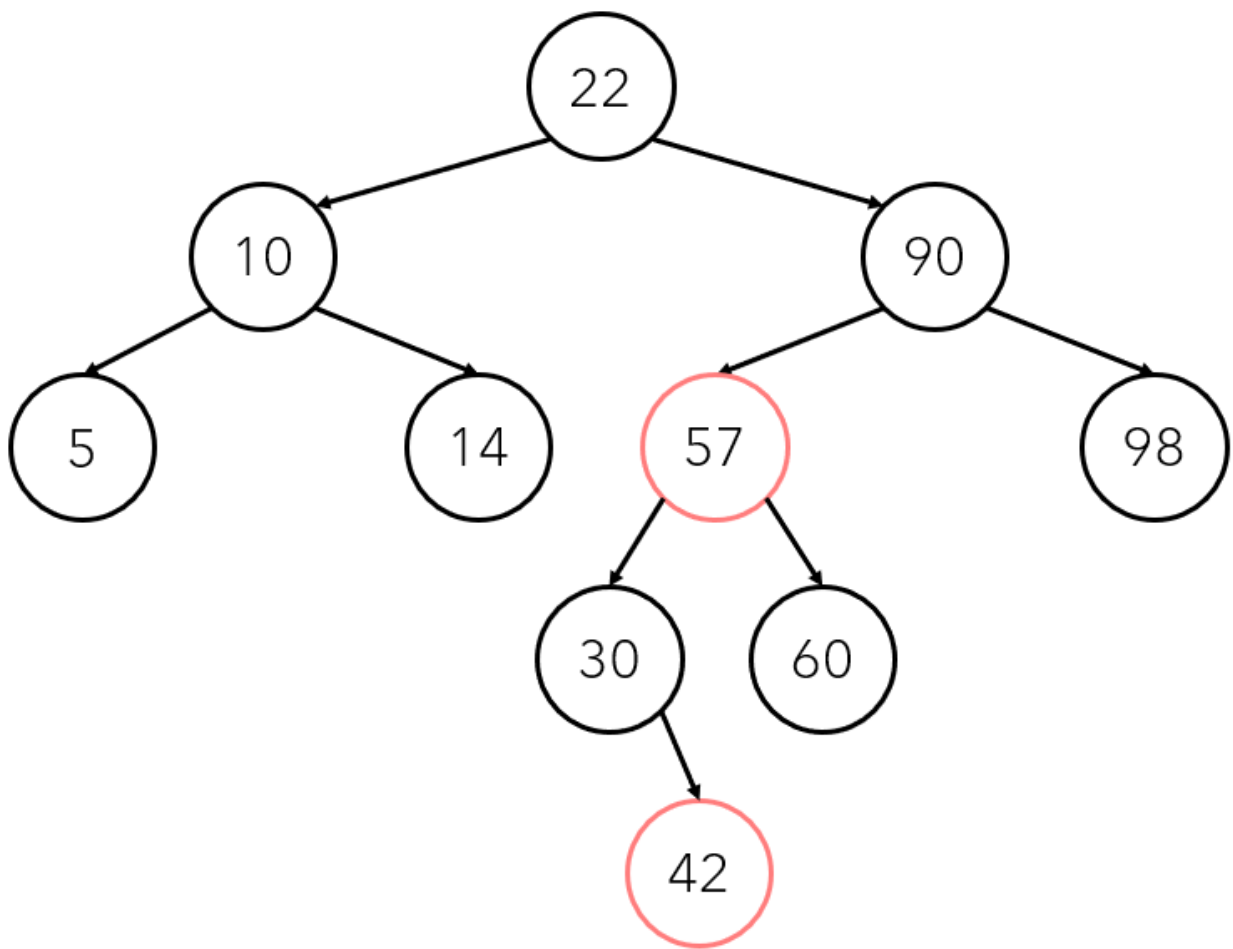
✓ + 1 pt 1 or 2

+ 0 pts Incorrect

(no title)

3 / 4 pts

+ 4 pts



+ 3.75 pts Very minor errors

+ 3.5 pts Minor errors

✓ + 3 pts Some errors

+ 2.5 pts Major errors

+ 2 pts Major errors

+ 1.5 pts Major errors

+ 1 pt Major errors

+ 0 pts Incorrect

Question 7

(no title)

1 / 1 pt

✓ + 1 pt a

+ 0 pts Incorrect

Question 8

(no title)

2.6667 / 3 pts

+ 3 pts

Index	Key
0	
1	15 → 8 → 1
2	
3	
4	
5	12 → 5
6	

✓ + 2.667 pts Inserted at the tail

+ 1 pt Probed

+ 0 pts Incorrect

Question 9

(no title)

1 / 1 pt

✓ + 1 pt 5/7

+ 0 pts Incorrect

Question 10

(no title)

3.2 / 4 pts

+ 4 pts

Index	Key
0	
1	1
2	15
3	
4	8
5	5
6	12

✓ + 3.2 pts 1 incorrect

+ 2 pts Major errors

+ 0 pts Incorrect

Question 11

(no title)

1 / 1 pt

✓ + 1 pt 5/7

+ 0 pts Incorrect

Question 12

(no title)

1 / 1 pt

✓ + 1 pt b

+ 0 pts Incorrect

Question 13

(no title)

1 / 1 pt

✓ + 1 pt a

+ 0 pts Incorrect

+ 4.5 pts

	Heaps (resizing)		Balanced BSTs		Hash Tables (resizing)	
Insert	$O(\log n)$	$O(n)$	$O(\log n)$	$O(\log n)$	$O(1)$	$O(n)$
Search	$O(n)$	$O(n)$	$O(\log n)$	$O(\log n)$	$O(1)$	$O(n)$
DeleteAS	$O(\log n)$	$O(n)$	$O(\log n)$	$O(\log n)$	$O(1)$	$O(n)$

+ 4.25 pts 1 incorrect

+ 4 pts 2 incorrect

✓ + 3.5 pts 4 incorrect

+ 2.75 pts 7 incorrect

+ 2 pts 10 incorrect

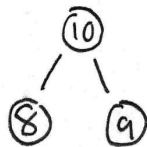
+ 0 pts Incorrect

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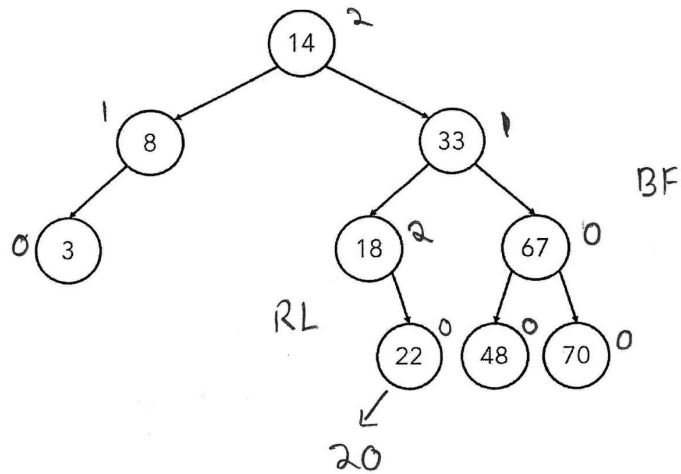
Name: Steven Chen

1. (5 points) Write basic pseudocode for BuildHeap that builds a min heap. You can call the bubbleDown helper method (you do not have to write out the code for bubbling down).

```
public static void buildHeap(List<Integer> list) {  
    Search parent.min.left_subTree  
    for i = current node index, i < 0, i-- do {  
        current = parent node of min.left_subTree  
        if (current < parent.current) {  
            bubbleDown {  
                i--  
            }  
        }  
    }  
}
```



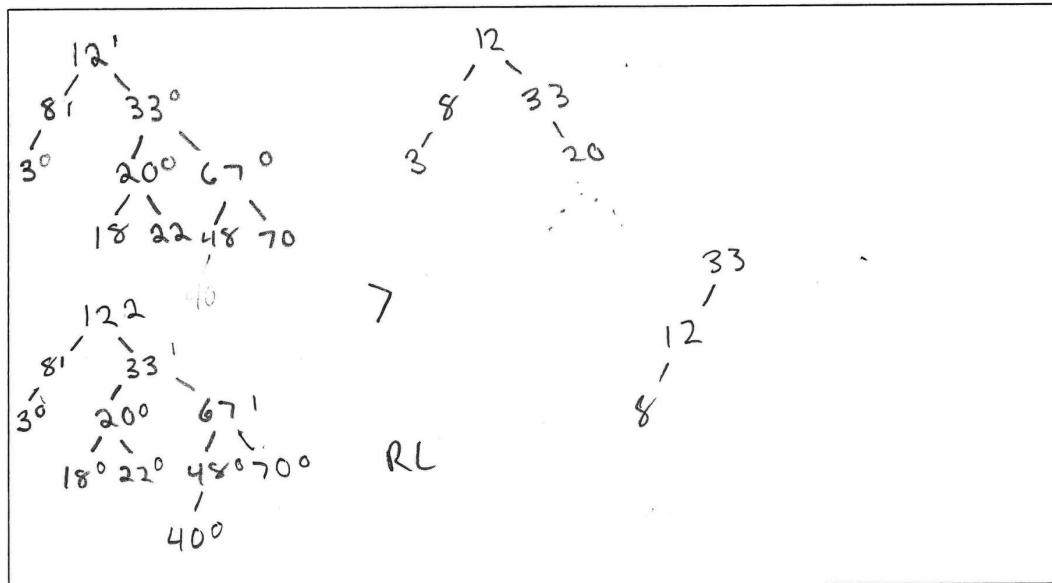
Consider the following AVL tree.



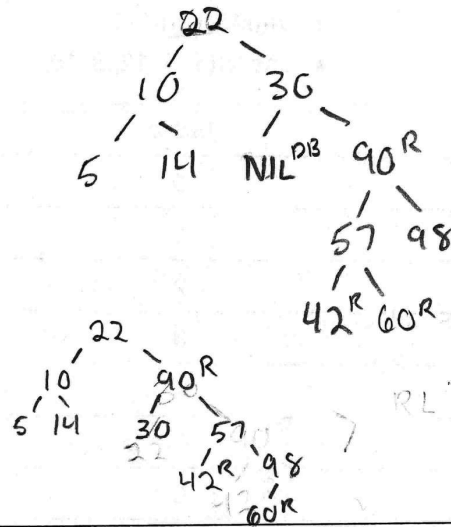
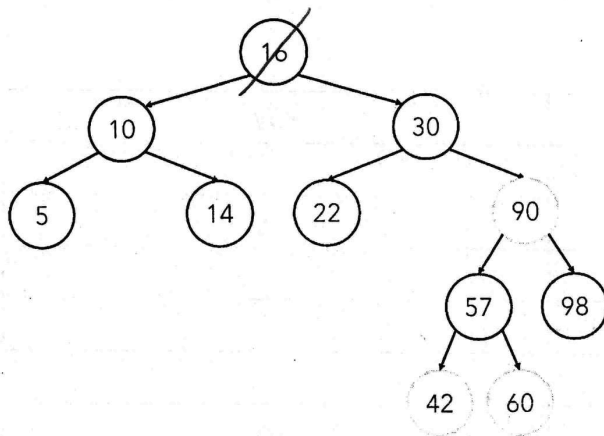
2. (1 point) If I insert 20, at which node does the first imbalance appear? Write "none" if there is no imbalance.

18

3. (3 points) Draw the resulting tree after 40 is inserted.



Consider the following red-black tree.



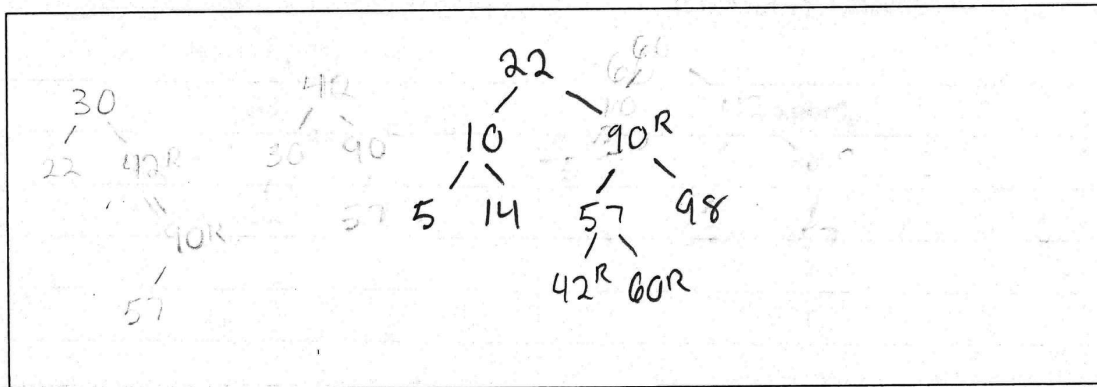
4. (1 point) What is the black height of this tree?

2

5. (1 point) What is the black height of node 30? Hint: Do not include node 30.

1

6. (4 points) Draw the resulting tree after 16 is deleted.



7. (1 point) Red-black trees are used more commonly than AVL trees because they are faster.

- a. True
- b. False

8. (3 points) Fill out the resulting chaining hash table that inserts at the head:

- $h_1(k) = k \bmod 7$
- insert(1, 5, 12, 8, 15)

Index	Key
0	
1	1 → 8 → 15
2	
3	
4	
5	5 → 12
6	

9. (1 point) What is the load factor of the above table? 5/7

10. (4 points) Fill out the resulting double hashing table (h_1k and insert are the same as above):

- $h_1(k) = k \bmod 7$
- $h_2(k) = 1 + (k \bmod 3)$
- $h(i, k) = (h_1(k) + i * h_2(k)) \bmod 7$
- insert(1, 5, 12, 8, 15)

$$\begin{aligned}
 1 \bmod 7 &= 1 \\
 5 \bmod 7 &= 5 \\
 12 \bmod 7 &= 5 \\
 1 \cdot (1 + 12 \bmod 3) &= (1 + 5) \bmod 7 \\
 1 + (1 \cdot 2) \bmod 7 &= 3 \\
 1 + (1 \cdot 1) \bmod 7 &= 2
 \end{aligned}$$

Index	Key
0	
1	1
2	15
3	8
4	
5	5
6	12

11. (1 point) What is the load factor of the above table? 5/7

12. (1 point) In a chaining hash table, the load factor tells us the

- a. Max chain length
- ☒ b. Average chain length
- c. Probability of collisions
- d. Fraction of table space used
- e. Fraction of table space still available
- f. Number of insertions until the next table resizing

13. (1 point) In a chaining hash table, a table with load factor greater than 1 is

- ☒ a. Possible because each array slot can store a chain of multiple keys
- b. Only possible if there are no collisions
- c. Impossible
- d. Always an error

14. (9 points) Fill out the time complexity table. Put amortized on the left and worst case on the right.

	Heaps (resizing)		Balanced BSTs		Hash Tables (resizing)	
Insert	$O(n)$	$O(n)$	$\log(n)$	$\log(n)$	$O(1)$	$O(n)$
Search	$O(n)$	$O(n)$	$\log(n)$	$\log(n)$	$O(1)$	$O(n)$
DeleteAS	$O(1)$	$O(1)$	$O(1)$	$\log(n)$	$O(1)$	$O(n)$



on your last data structures quiz ever !!!1!!11!!!!1!!

